

Maths Homework Template

Author

September 19, 2022

Short Math Guide: <http://tug.ctan.org/info/short-math-guide>.

Short Math Guide (Chinese version): <https://github.com/hoganbin/short-math-guide>.

- ¶ Inline equations for simple math: $\boldsymbol{p} = \hbar \boldsymbol{k}$, $\varepsilon = \hbar \omega$.

¶ Centered equations for more involved or important equations:

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{i=1}^n a_i b_i \quad (1)$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \mathrm{e}^{a(x)} = a'(x) \mathrm{e}^{a(x)} \quad (2)$$

$$\frac{\|\boldsymbol{A}\boldsymbol{x}\|_{\infty}}{\|\boldsymbol{x}\|_{\infty}} = \frac{\max_i \left| \sum_{j=1}^n A_{ij} x_{j1} \right|}{\max_i |x_{i1}|} \quad (3)$$

¶ An example of a matrix:

$$\boldsymbol{M} = \begin{bmatrix} \frac{1 + \sqrt{5}i}{2} & \mathrm{e}^{-i\omega t} \\ i\mathrm{e}^{-i\omega t} & \frac{1 - \sqrt{5}i}{2} \end{bmatrix} \quad (4)$$

¶ An example of intergals:

$$\iiint \nabla \cdot \boldsymbol{A} \, \mathrm{d}V = \oiint \boldsymbol{A} \, \mathrm{d}\boldsymbol{S} \quad (5)$$

¶ An example of roots:

$$r = \sqrt[3]{\frac{3}{4\pi N}} \quad (6)$$

¶ An example of math fonts:

$$\begin{array}{cccc} \text{ABCD} & \textit{ABCD} & \boldsymbol{ABCD} & \textit{ABCD} \\ \mathcal{ABCD} & \mathcal{A}\mathcal{B}\mathcal{C}\mathcal{D} & \mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} & \text{ABCD} \end{array} \quad (7)$$

¶ An example of Dirac brackets:

$$\begin{array}{ccc} \langle \phi | \psi \rangle & \left\langle \frac{1}{2} \lambda \middle| \psi \right\rangle & |\phi_{\hat{Q}=0}\rangle \\ \langle A \rangle & \left\langle \frac{1}{2} \lambda \right\rangle & \langle \Psi | A | \Psi \rangle \end{array} \quad (8)$$

2.

$$\begin{aligned} I(x) &= \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathrm{d}I(x) \\ &= 4\alpha \frac{I_0}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} \cos^2 \left(\frac{\pi d}{\lambda D} \left(x + \frac{D}{R} \xi \right) \right) \mathrm{d}\xi \\ &= 4\alpha I_0 \left(1 + \frac{1}{u} \sin u \cos \left(\frac{2\pi d}{\lambda D} x \right) \right), \quad u = \frac{\pi db}{\lambda R} \end{aligned}$$

3.

$$F_n(z) = \begin{cases} \frac{1}{\sqrt{1 + in^2 \mathrm{e}^{Az}}}, & z < 0 \\ \frac{1}{\sqrt{1 + n^2 \mathrm{e}^{Az}}}, & z \geq 0 \end{cases}$$

4.

$$u(x,t)=\sum_{n=0}^{\infty}X_n(x)Y_n(y)=\frac{8a^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\left(1-\frac{\cosh\frac{2k+1}{a}\pi y}{\cosh\frac{2k+1}{2a}\pi b}\right)\sin\frac{2k+1}{a}\pi x$$

5.

$$\begin{aligned} \mathcal{L}\left\{\frac{1-\cos\omega t}{t}\right\} &= \int_p^\infty \left(\frac{1}{p}-\frac{1}{2(p+\mathrm{i}\omega)}-\frac{1}{2(p-\mathrm{i}\omega)}\right)\mathrm{d}t \\ &= \ln p - \frac{1}{2}\ln(p+\mathrm{i}\omega) - \frac{1}{2}\ln(p-\mathrm{i}\omega)\Big|_p^\infty \\ &= \frac{1}{2}\ln\frac{p^2+\omega^2}{p^2} \end{aligned}$$

6.

$$\begin{aligned} [q,p]_{Q,P} &= \frac{\partial q}{\partial Q} \cdot \frac{\partial q}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial q}{\partial Q} \\ &= 1 + \frac{P^2 \mathrm{e}^{2Q}}{1 - P^2 \mathrm{e}^{2Q}} - \frac{P^2 \mathrm{e}^{2Q}}{1 - P^2 \mathrm{e}^{2Q}} \\ &= 1 \end{aligned}$$

7.

$$\begin{aligned} \frac{4\pi R_\odot^2\cdot\sigma T_\odot^4}{4\pi d^2} &= 1352.83\,\mathrm{W/m^2} \\ T_\odot &= 5763.43\,\mathrm{K} \\ \lambda_\odot &= \frac{b}{T_\odot} = 5027.86\,\mathrm{\AA} \end{aligned}$$